



2687 - Leveraging Early Public JWST Data to Measure Luminosity Functions and Rest-UV Slopes from $6 < z < 12$

Cycle: 1, Proposal Category: AR

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OBSERVATIONS

ABSTRACT

The evolution of the rest-frame ultraviolet (UV) luminosity function encodes information on the key physical properties regulating galaxy evolution. While theoretical models, invoking a variety of sub-grid models, are able to match observations at $z < 8$, their predictions have significant scatter at higher redshifts. This is understandable at $z > 9$, Hubble only probes galaxies with a single imaging filter, so there is a variety of tension between observational studies. This will change with Webb; Deep near-infrared imaging planned early in Cycle 1 will probe 1-5 μ m, allowing the selection of large samples of robust $z \sim 10$ galaxies. We propose for archival funding to analyze the data from four publicly available datasets (two Early Release Science programs, and two Guaranteed Time Observer programs with public releases). We will analyze these data in a systematic way to select robust samples of galaxies in this epoch, to 1) measure the shape of the bright end of the rest-UV luminosity function, constraining models of star-formation efficiencies, and 2) measure robust rest-UV colors, placing limits on the chemical enrichment in early galaxies. Doing these analyses early in the Webb mission will allow our first glimpse into the physical processes dominating galaxy evolution in the first 500 Myr, setting the stage for future Cycle 2 programs to push farther into the early universe.

OBSERVING DESCRIPTION

We will use NIRCcam imaging from two ERS programs (PIDs 1324 and 1345) as well as the portions of two GTO programs (PIDs 1176 and 1199) that will be publicly available immediately upon observation in Cycle 1. The combined imaging from all four programs will cover ~ 135 arcmin², with a dynamic range spanning 1 magnitude in any given band. At a minimum, all four programs will obtain imaging over 1-5 micron using at depth greater or equal to 28.9 (for a 5point source) in F150W. We describe the datasets below.

PID 1176 (PI Windhorst) - JWST Medium-Deep Fields: The first epoch of the NIRCcam imaging of the JWST North Ecliptic Pole Time-Domain Field (NEP TDF), representing one of the field ‘spokes’. The imaging will reach 5sigma depths of ~ 28.3 -28.9 (measured for a resolved source) across a total area of ~ 12.1 arcmin².

Filter coverage: F090W, F115W, F200W, F277W, F356W, F410M, F444W

PID 1199 (PI Stiavelli) - The metallicity of galaxies in the MACS J1149.5+2223 field: One pointing of NIRCcam imaging in parallel, covering 9.2

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arcmin² at a depth of $m \sim 29.1$. For the current version of the APT for this program, 50% of the NIRCcam FOV overlaps with the HFF parallel for the MACS J1149.5+2223 field, making deep HST imaging also available for our analysis in this field.

Filter coverage: F115W, F200W, F277W, F356W, F410M, F444W

PID 1324 (PI Treu) - Through the Looking GLASS: We will include the two NIRCcam pointings obtained in parallel, which will reach depths of ~ 29.0 and ~ 28.7 for a combined area of ~ 18.4 arcmin². A small range of PAs will result in overlap with the Abell 2744 HFF parallel field, providing another dataset partially covered by deep HST imaging.

Filter coverage: F090W, F115W, F200W, F277W, F356W, F444W

ID 1345 (PI Finkelstein) - Cosmic Evolution Early Release Science (CEERS): The CEERS NIRCcam imaging will cover ~ 87.1 arcmin² to a depth $m \sim 28.4$. The CEERS NIRCcam imaging will be fully covered by HST imaging to a depth of $m \sim 27.6$.

Filter coverage: F115W, F200W, F277W, F356W, F410M, F444